

Fungal Infections of the Urinary Tract

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Funguria is a frequent finding in hospitalized patients. The organisms found in urine are almost always *Candida* spp., although several other yeasts and, less often, molds and endemic fungi, can be found (Table 55.1). Candiduria is not a symptom, a sign, or a disease, but frequently it is a perplexing phenomenon for the physician to address. Most patients with candiduria are asymptomatic and have colonization of the bladder or an indwelling urinary catheter. The most difficult diagnostic problem is determining when infection, rather than colonization, is present. Diagnostic tests to define colonization or infection have not been standardized, and neither have studies to localize the site of infection to either the bladder or the kidneys. In contrast to the situation with candiduria, growth in urine of organisms such as *Blastomyces dermatitidis*, *Aspergillus* spp., and *Cryptococcus neoformans* almost always reflects disseminated infection. This chapter outlines an approach to the diagnosis and treatment of candiduria and other less common fungal urinary tract infections (UTIs).

CANDIDA

Epidemiology

Candida spp. are common inhabitants of the perineum but are not found in urine in appreciable numbers in healthy hosts. A variety of predisposing factors allow these commensals microorganism to grow in the urine and in some cases to cause infection of the bladder or the upper urinary tract. These factors are frequently encountered in hospitalized patients, especially those in the intensive care unit (ICU).¹ In a cross-sectional survey of positive urine cultures obtained from hospitalized patients, *Candida* spp. were found in almost 10% of specimens and were the third most common microorganism isolated from urine.²

Risk factors for candiduria, but not specifically for *Candida* UTI, include increased age, female sex, antibiotic use, urinary drainage devices, prior surgical procedures, and diabetes mellitus^{3,4} (Table 55.2). In a large surveillance study, urinary drainage devices, mostly indwelling urethral catheters, were present in 83% of 861 patients who had candiduria.³ In a multicenter study of patients in an ICU, independent risk factors associated with candiduria were older than 65 years, female sex, diabetes mellitus, prior antibiotic use, mechanical ventilation, parenteral nutrition, and length of hospital stay before ICU admission.⁴ Candiduria uncommonly leads to candidemia, confirming that most often candiduria reflects colonization, not infection.³ When candiduria is found in patients with candidemia, the strains often have been shown to be unrelated by molecular techniques.⁵ Prospective controlled studies assessing risk factors for well-documented *Candida* UTI have not been performed because firm diagnostic criteria specifically for infection have not been established. However, clinical experience suggests that UTI is more common in diabetic patients and in those with urinary tract obstruction.

Pathogenesis

Candida can cause urinary tract disease by either the hematogenous or the ascending route. The pathogenesis of hematogenous seeding of *Candida* to the kidney has been studied with animal models.⁶ Multiple microabscesses develop throughout the cortex, with the yeasts penetrating through the glomeruli into the proximal tubules, where they are shed into the urine (Fig. 55.1). Healthy animals eventually clear the infection, but immunocompromised animals do not. Consistent with these studies, kidney microabscesses have been identified at autopsy in many patients with candidemia or invasive candidiasis. For ascending infection with *Candida*, obstruction is an important factor in many patients. Virulence factors that control adherence and biofilm formation are likely important.^{6,7}

A unique syndrome seen early after kidney transplantation is graft site candidiasis, which appears to result from contamination of the donor kidney during the harvest procedure.⁸ Arteritis with aneurysm formation and rupture can result from direct fungal invasion into the arterial wall. Most patients lose the graft, and mortality is high.

Microbiology

C. albicans accounts for 50% to 70% of all *Candida* urinary isolates, and *C. glabrata* for about 20% of isolates.³ *Candida tropicalis* and *Candida parapsilosis* are less common, and other species are rarely isolated. Certain populations of patients have a predominance of *C. glabrata* isolated from cultures of urine; these include older adults, those who have diabetes, kidney transplant recipients, and persons who have had prior exposure to fluconazole. To prescribe appropriate treatment it is important to know the species causing candiduria. Resistance to fluconazole, the primary agent used for the treatment of *Candida* UTI, is common among isolates of *C. glabrata* and in all isolates of *Candida krusei*. In contrast, almost all isolates of *C. albicans*, *C. tropicalis*, and *C. parapsilosis* are susceptible to fluconazole.

Clinical Manifestations

Most patients with candiduria are asymptomatic, and, indeed, most do not have infection. A large prospective surveillance study of patients with candiduria noted that less than 5% of patients with candiduria had any symptoms suggesting UTI.³ When patients have symptomatic cystitis or pyelonephritis, symptoms are indistinguishable from those noted with bacterial infections. Cystitis is manifested by dysuria, frequency, urgency, and suprapubic discomfort; patients with upper tract infection can present with fever, chills, and flank pain. Urinary tract obstruction can occur from formation of a bezoar or fungal ball in the bladder or the collecting system.

Patients who have had seeding of the kidney parenchyma during an episode of candidemia manifest the symptoms associated with invasive candidiasis, such as chills, fever, and hypotension, and not symptoms suggesting UTI.

TABLE 55.1 Fungal Genitourinary Tract Infections

Fungal Infection	Prostate	Bladder	Kidney
Candidiasis	++	++++	++++
Cryptococcosis	++	+/-	+++
Blastomycosis	+++	+	++
Histoplasmosis	+	+/-	++
Coccidioidomycosis	+	+/-	++
Aspergillosis	+/-	+/-	++
Mucormycosis	+/-	+/-	++

Shown is the relative frequency of the site of infection for various fungal organisms.

TABLE 55.2 Risk Factors for Candiduria

Type	Risk Factors
Kidney (hematogenous)	Neutropenia, recent surgery, central venous catheter, parenteral nutrition, antibiotics, dialysis
Lower urinary tract	Indwelling bladder catheter, older age, female, diabetes, obstruction, antibiotics, urinary tract instrumentation
Upper urinary tract	Older age, diabetes, antibiotics, obstruction, urinary tract instrumentation (e.g., nephrostomy tube, ureteral stent)

Diagnosis

Major diagnostic difficulties are encountered in trying to differentiate contamination of a urine specimen from colonization of the bladder or an indwelling urethral catheter from invasive infection of the bladder or kidney. Contamination is most easily differentiated by simply repeating the urine culture a day later to determine whether candiduria persists. It may be necessary to obtain the second urine specimen by sterile bladder catheterization if the patient is unable to accomplish a clean-catch collection. In those patients who have an indwelling urethral catheter, the catheter should be replaced and a second urine specimen collected the next day. For either of these circumstances, if the repeated culture yields no yeasts, no further diagnostic studies or therapeutic interventions are needed.

Distinguishing colonization from infection is not straightforward. Compared with bacterial UTIs, in which the diagnosis is based on appropriate symptoms combined with the findings of pyuria and quantitative bacterial counts, no studies have established the importance of quantitative urine cultures or pyuria for the diagnosis of *Candida* UTI.

The role of quantitative urine cultures to differentiate upper tract infection from bladder colonization was assessed by investigators in the 1970s and unfortunately showed broad ranges of colony counts for both colonization and infection.⁹ In patients who did not have indwelling catheters, documented kidney infection was found with colony counts as low as 10^4 yeast colony-forming units per milliliter (cfu/mL). For patients who had indwelling catheters, colony counts between 2×10^4 and 10^5 cfu/mL or more were noted and there was no correlation with biopsy-proven kidney infection.

The techniques routinely used in most clinical laboratories for the detection of bacteria will also detect *Candida* in urine. However, *C. glabrata* grows more slowly than other species, and colonies may not appear for 48 hours, which is often after routine cultures of urine have been discarded. Laboratories should be notified if *C. glabrata* is a likely pathogen.

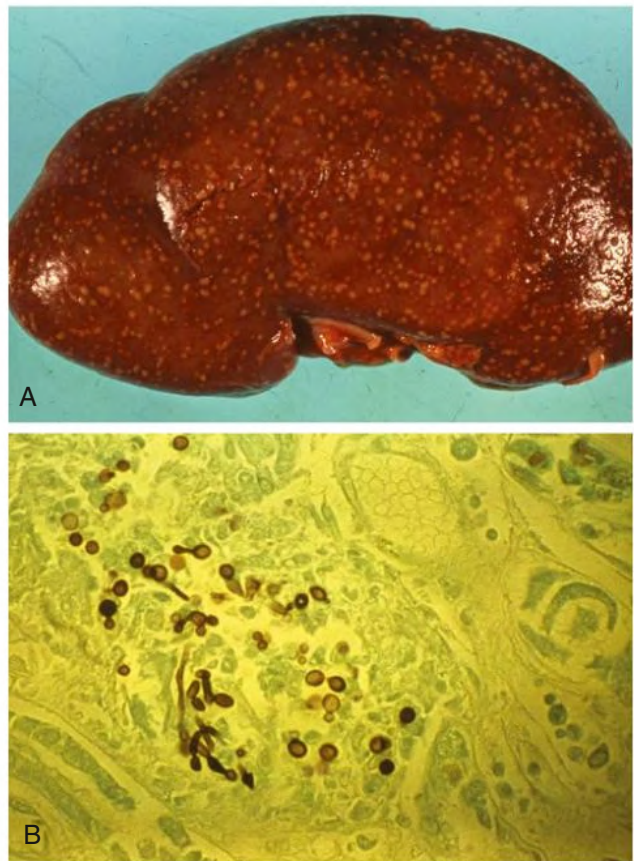


Fig. 55.1 Hematogenous Candidiasis Involving the Kidney. (A) Multiple small abscesses are seen throughout the kidney. (B) Histopathologic demonstration of a microabscess caused by *Candida albicans* in the cortex of the kidney (methenamine silver stain; yeast shown in gray-brown; magnification $\times 100$).

Pyuria is often not a helpful diagnostic criterion for infection in patients with candiduria. Concomitant bacteriuria, frequently noted in patients with candiduria, may be responsible for pyuria, and when an indwelling bladder catheter is present, pyuria is routinely noted. In patients who do not have an indwelling bladder catheter or bacteriuria, the presence of pyuria is helpful.

Imaging procedures, including abdominal ultrasound and computed tomography (CT), are essential to document obstruction at any level in the urinary tract and to determine the presence of fungus balls in bladder or kidney (Fig. 55.2). In some patients, it is helpful to perform cystoscopy and biopsy of the bladder wall to determine whether inflammation is present and evaluate the extent of invasion (Fig. 55.3).

Treatment With Systemic Antifungal Agents

Most patients who have candiduria do not need treatment with an antifungal agent. For patients who are asymptomatic, treatment should be given only to those who are at high risk for development of candidemia or in whom the presence of candiduria, regardless of symptoms, is likely to represent disseminated infection. The guidelines for the management of candidiasis published by the Infectious Diseases Society of America recommend treatment for patients about to undergo urologic procedures, infants with very low birth weight, and neutropenic patients¹⁰ (Table 55.3). Patients who have candiduria and who are to undergo a urologic procedure are at increased risk for development of candidemia and should be treated with an antifungal agent a few days before and after the procedure. Candiduria in



Fig. 55.2 Fungus Balls in the Pelvis of the Kidney. Several fungus balls (dark round “holes” in the contrast dye) are seen in the ureter and pelvis of the right kidney, causing hydronephrosis.

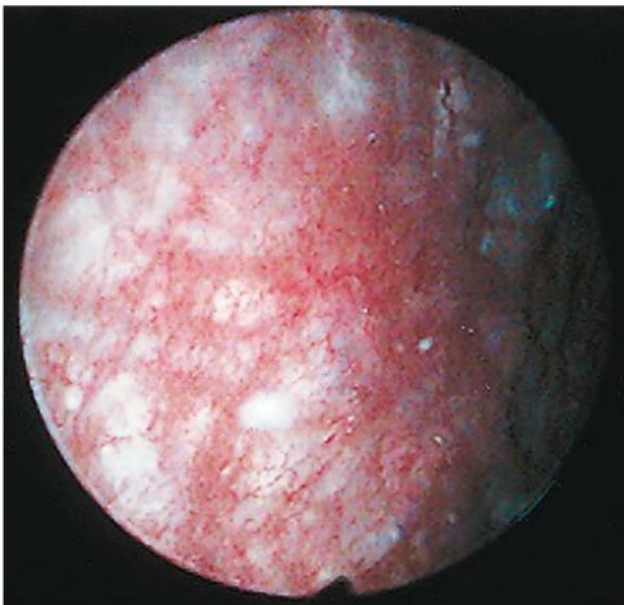


Fig. 55.3 Candida Cystitis. Cystoscopic appearance of extensive cystitis caused by *Candida krusei*.

neutropenic patients and in infants with very low birth weight has a high probability of representing disseminated candidiasis; thus these groups should be treated with an antifungal drug. Asymptomatic candiduria in the kidney transplant patient does not warrant systemic antifungal treatment unless obstruction is present; a urologic procedure, including stent removal, is planned; or symptoms suggestive of local or systemic infection develop.^{10,11}

In non-high-risk patients who have asymptomatic candiduria, removal of an indwelling urinary catheter will eradicate candiduria in many patients.³ If catheterization cannot be discontinued, the existing catheter should be removed and a new one inserted. This will often eradicate candiduria transiently, but it is highly likely that the organisms will return within a short time. It is not known if a suprapubic catheter is less likely to be colonized than an indwelling urethral catheter. Relief of obstruction, whether it is present in the upper or lower

urinary tract, is essential for the long-term eradication of *Candida* from the urinary tract.

Patients who have symptoms suggestive of cystitis or pyelonephritis, and in whom bacteria and *Candida* are found in the urine culture specimen, should be treated initially with an antibacterial agent. If no bacteria are present, treatment with an antifungal drug is appropriate. Eradication of the organism with antifungal therapy is more likely if the indwelling catheter is removed.¹² Oral fluconazole, which is an azole antifungal agent and is excreted as active drug in the urine, is the agent of choice (see Table 55.3). A loading dose of 400 mg should be given, followed by 200 mg daily for 14 days.^{10,13} Fluconazole does not effectively treat *C. krusei* infections, and many *C. glabrata* infections also do not respond to fluconazole.

The possibility of drug-drug interactions should be evaluated before fluconazole is prescribed. Phenytoin, warfarin, cyclosporine, tacrolimus, and sulfonylurea agents are a few of the drugs for which serum concentrations will increase and may reach toxic levels after the addition of fluconazole. Azoles should be used with caution in patients taking drugs that prolong the QTc interval.

The other available azole agents, itraconazole, voriconazole, posaconazole, and isavuconazole, are not excreted into the urine as active drug. Whether tissue concentrations might be high enough to treat invasive kidney or bladder infections is not known, but there is little clinical experience to suggest that they will be effective.

Intravenous (IV) amphotericin B deoxycholate is effective in treating *Candida* UTI but should be reserved for patients who have upper tract infection or for whom fluconazole therapy has failed, which will be predominantly those who have *C. glabrata* infection. Because of its inherent nephrotoxicity, IV amphotericin B deoxycholate must be used judiciously in patients who have kidney dysfunction. The recommended dosage is 0.3 to 0.6 mg/kg/day for 1 to 7 days, although this may be extended to 2 weeks for those who have complicated upper tract infection.¹⁰ The dose used depends on kidney function and number of days infusions will be given; for example, some clinicians have had success with 0.3 mg/kg/day for 3 days, whereas others use a single dose of 0.6 mg/kg.¹³ Infusion-related side effects are seen in some patients, even when low doses are used, and can include rigors, fever, nausea, vomiting, and headache.

TABLE 55.3 Treatment Recommendations for Candiduria and *Candida* Urinary Tract Infections

Infection	Treatment	Other/Alternative Therapy
Asymptomatic candiduria	Urologic surgery	Treat a few days before and after the procedure with fluconazole 200–400 mg/day PO or amphotericin B deoxycholate 0.3–0.6 mg/kg/day IV
	Low-birthweight infant	Treat as for disseminated candidiasis/candidemia with fluconazole 12 mg/kg/day
	Neutropenic patient	Treat as for disseminated candidiasis/candidemia with an echinocandin
Cystitis	Preferred: fluconazole 200 mg/day PO × 14 days ^a	Alternatives include amphotericin B deoxycholate 0.3–0.6 mg/kg/day IV × 1–7 days or flucytosine 25 mg/kg q6h PO × 7–10 days ^b
Pyelonephritis	Preferred: fluconazole 200–400 mg/day PO × 14 days	Alternatives include amphotericin B deoxycholate 0.3–0.6 mg/kg/day IV × 1–7 days with or without flucytosine 25 mg/kg q6h PO or flucytosine 25 mg/kg q6h PO × 14 days
Kidney (hematogenous)	Treat as for disseminated candidiasis/candidemia with an echinocandin or fluconazole	—
Fungus balls	Surgical removal <i>plus</i> fluconazole 200–400 mg/day PO until resolved	Alternatives include surgical removal <i>plus</i> amphotericin B deoxycholate 0.3–0.6 mg/kg/day IV with/without flucytosine 25 mg/kg q6h PO Irrigation through nephrostomy tube with amphotericin B deoxycholate 25–50 mg in 200–500 mL sterile water
Prostatitis Epididymo-orchitis	Surgical drainage <i>plus</i> fluconazole 400 mg/day PO until resolution noted on imaging studies	Alternatives include surgical drainage <i>plus</i> amphotericin B deoxycholate 0.3–0.6 mg/kg/day IV

^aFluconazole dosage in kidney failure: creatinine clearance (CrCl) 20 to 50 mL/min, reduce dose by 50%; CrCl <20 mL/min, reduce dose by 75%.

^bFlucytosine dosage in kidney failure: CrCl 20 to 40 mL/min, reduce dose to 25 mg/kg q12h; CrCl <20 mL/min, reduce dose to 25 mg/kg q24h.

IV, Intravenous; PO, orally; q6h, every 6 hours.

Lipid formulations of amphotericin B are not recommended for treatment of fungal UTI. The decreased nephrotoxicity that results from the addition of the lipid component likely also precludes the drug's effectiveness by failing to achieve adequate levels in the urinary tract.¹³

One of the few uses of flucytosine is for the treatment of *Candida* UTI. Flucytosine is excreted into the urine in high concentrations but should be used only when fluconazole is not tolerated or the organism is fluconazole resistant. The usual dosage of flucytosine in patients who have normal creatinine clearance is 25 mg/kg orally every 6 hours for 7 to 10 days.¹³ Most species of *Candida*, with the exception of *C. krusei*, are susceptible to flucytosine, but resistance emerges quickly when this agent is used alone. Serious adverse effects of flucytosine include bone marrow suppression and hepatotoxicity. These effects are dose related, and the risk increases greatly with kidney failure (see Table 55.3).

The echinocandins (caspofungin, micafungin, and anidulafungin) have minimal or no excretion into the urine as active drug. The tissue concentrations achieved with these agents may be adequate to treat invasive *Candida* infections of the bladder or kidney, but this is not clear. Both success and failure have been reported when echinocandins are used to treat *Candida* UTI.^{14–16} Currently, echinocandins are not recommended for the treatment of *Candida* UTI but can be tried if all else fails in patient with *C. glabrata* UTI.

Local Antifungal Administration

Continuous bladder infusion of amphotericin B deoxycholate 50 mg in 1 L of sterile water through a triple-lumen catheter is sometimes used to treat *Candida* bladder infection. Bladder irrigation clears candiduria more quickly than systemic antifungal agents.¹⁷ However, the effect is brief, and recolonization occurs within 1 to 2 weeks. Bladder irrigation is rarely ordered, but it is sometimes useful for patients who have recalcitrant bladder infection with azole-resistant *C. krusei* or *C. glabrata*.

In the patient with kidney obstruction caused by a fungus ball, irrigation through a percutaneous nephrostomy tube with amphotericin B deoxycholate is recommended, in addition to systemic antifungal

therapy with fluconazole.¹⁰ Absorption of amphotericin B does not occur, and direct infusion is not nephrotoxic. Surgical or endoscopic removal of the fungus ball is essential to eradicate the infection.

Localized *Candida* Infections

Prostatitis and prostatic abscess caused by *Candida* spp. manifest with symptoms that are similar to those of bacterial prostatic infection. The initial manifestation may be urinary retention; physical examination reveals a tender prostate, and imaging can show either discrete abscesses or diffuse inflammation. Treatment is drainage, if an abscess is present, and fluconazole, which achieves excellent concentrations in the prostate, for several months until the infection has resolved.¹⁸ Epididymo-orchitis is less common and usually manifests as a tender scrotal mass. Surgical drainage or orchiectomy is required, along with fluconazole therapy until resolution has occurred.

OTHER YEASTS

Rarely, yeasts such as *Saccharomyces cerevisiae* and *Trichosporon asahii* cause UTI, but the most important non-*Candida* yeast that infects the genitourinary (GU) tract is *Cryptococcus neoformans*. Cryptococcosis occurs primarily in immunosuppressed hosts. In autopsy series, kidney involvement has been noted in 25% to 50% of patients who died of cryptococcosis, but GU tract symptoms were rare. The prostate is frequently infected and can be a reservoir for persistent *C. neoformans* infection. Isolated prostatitis and epididymo-orchitis have been reported in the absence of systemic cryptococcosis.¹⁹ The diagnosis is usually made after biopsy of a mass or nodule; granulomatous inflammation is typically seen. Treatment of localized GU tract cryptococcal infection is fluconazole 400 mg/day for 6 to 12 months.

ASPERGILLUS AND OTHER MOLDS

The urinary tract is an uncommon site of infection with molds. However, individual case reports have noted GU infections caused by a variety of molds, including the Mucorales (e.g., *Rhizopus*,

Mucor) and *Aspergillus*.^{20,21} The most common mold infection is aspergillosis. Hematogenous spread to the kidney with invasive disease in immunosuppressed patients results in numerous kidney microabscesses and infarcts. This may be an incidental finding at autopsy in patients with disseminated infection; in other patients, infection is localized to the GU tract.²¹ Urinary tract obstruction from masses of fungal elements is not uncommon. Treatment is surgical removal of the obstructing mass, often nephrectomy, and systemic antifungal therapy for the specific mold. Mortality is extremely high when GU involvement is accompanied by disseminated infection.

ENDEMIC FUNGI

All of the major endemic mycoses have been reported to infect the GU tract. For all of these organisms, infection is by hematogenous spread to the GU tract. For upper tract infection, treatment is the same as that for disseminated infection. The treatment of focal infection, which is more likely to involve the lower GU tract,

often requires surgical removal of the infected tissue and antifungal therapy.

B. dermatitidis has the greatest propensity to cause symptomatic infection. In patients with disseminated blastomycosis, involvement of the GU tract occurs in as many as a third of cases and usually manifests as prostate or epididymal infection.²² In most patients, this involvement is discovered incidentally when urine cultures yield the organism or a biopsy is performed for a prostatic or epididymal mass.

Symptomatic GU tract involvement with histoplasmosis is uncommon. However, at autopsy, kidney lesions are often found in patients who have disseminated histoplasmosis.²³ Patients are usually asymptomatic in regard to urinary symptoms. Individual cases of testicular abscesses, epididymitis, and prostate nodules have been reported.

Coccidioidomycosis rarely causes symptomatic UTI. However, autopsy series of cases of disseminated coccidioidomycosis have noted kidney involvement in more than 50% of cases, and *Coccidioides* spp. can be grown in urine culture in patients with disseminated infection.²⁴ Localized infection, manifesting as abscesses or mass lesions of the epididymis or prostate, has also been reported caused by *Coccidioides* spp.

SELF-ASSESSMENT QUESTIONS

- Which of the following *Candida* spp. is isolated *most* often in patients with candiduria?
 - C. glabrata*
 - C. parapsilosis*
 - C. albicans*
 - C. krusei*
 - C. tropicalis*
- The presence of pyuria is a helpful diagnostic test for *Candida* UTI when there is:
 - An indwelling bladder catheter in a symptomatic patient
 - An indwelling bladder catheter in an asymptomatic patient
 - Concomitant bacteriuria and candiduria in an asymptomatic patient
 - Concomitant bacteriuria and candiduria in a symptomatic patient
 - No indwelling bladder catheter and no bacteriuria
- The dose of which of the following antifungal agents used to treat fungal UTI should be reduced in patients who have a creatinine clearance below 40 to 50 mL/min?
 - Amphotericin B
 - Flucytosine
 - Caspofungin
 - Liposomal amphotericin B
 - Voriconazole
- The antifungal agent of choice for treating a *Candida albicans* UTI is:
 - Amphotericin B
 - Fluconazole
 - Flucytosine
 - Voriconazole
 - Caspofungin

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